Newton's laws & practical

Explain motion

1 First law & inertia

Zero resultant → rest or constant velocity. Inertia is the tendency to retain that motion.

Balanced forces do not mean no forces. A steady car has driving force = resistance.

2 Second & third laws

F = ma, using resultant F. Inertial mass = F/a. More mass → less a for same F.

Third law: equal opposite forces of same type on different objects. Weight and normal force on one book are not a pair.

Follow falling forces

3 Terminal velocity

Balanced forces → constant velocity

Initially weight > drag → downward acceleration. Speed rises → drag rises → resultant and a fall.

Drag = weight → constant terminal velocity. Opening parachute increases drag, slowing fall to a lower terminal speed.

4 Measure acceleration

Low-friction trolley, string, pulley, hanging mass. Light gates or motion sensor measure velocity changes.

Calculate $a = \Delta v/\Delta t$ before hanger hits floor. Secure pulley; stop trolley safely; control friction.

Measure acceleration

5 Vary force

Transfer masses from trolley to hanger: total moving mass constant, driving weight increases.

For whole system, driving force ≈ hanger weight if friction negligible. Trolley tension alone is not hanger weight while accelerating.

6 Vary mass

Keep hanging mass fixed; add mass to trolley. Force approximately constant, total mass increases.

Repeat; a vs F should be proportional at fixed mass. a vs 1/m should be proportional at fixed force.